



Lab 07 - Inheritance

Name your solution **Week07Inheritance**

A. Project Bank

Given this source code with an output like this

C++

Copy

```
#include <iostream> #include <vector> #include <format> // Requires C++20
class BankAccount { private: std::string _name; int _balance; public:
std::string name() const { return _name; } BankAccount() { _name = "";
_balance = 0; } BankAccount(std::string name, int balance) : _name(name),
_balance(balance) {} int withdraw(int amount) { if (_balance >= amount) {
_balance -= amount; return amount; } return 0; } }; auto getAllAccounts() {
return std::vector<BankAccount> { BankAccount("Alice", 60000),
BankAccount("Bob", 100000), BankAccount("Charlie", 100000),
BankAccount("David", 200000), BankAccount("Eve", 300000), }; } int main() {
auto accounts = getAllAccounts(); for (auto& account : accounts) { std::cout
<< std::format("{} can withdraw {} \n", account.name(),
account.withdraw(70000)); } }
```

Alice can withdraw 0 Bob can withdraw 70000 Charlie can withdraw 70000 David
can withdraw 70000 Eve can withdraw 70000

Requirements

1. Create a class **BankAccountV2** that inherits from the class **BankAccount**.
2. In the class **BankAccountV2**, we want to support the withdrawal with one business extra constraint: the account's balance after withdrawal must be greater than or equal 50.000.
3. Create another **getAllAccountsV2()** function, this time, return **BankAccountV2** objects instead of **BankAccount** objects.

Expected output

```
Alice can withdraw 0 Bob can withdraw 0 Charlie can withdraw 0 David can
withdraw 70000 Eve can withdraw 70000
```

B. Project Blog post

Given this source code with an output like this

```
class Post { private: std::string _author; std::string _title; std::string
_content; public: Post () {} Post (std::string author, std::string title,
std::string content) : _author(author), _title(title), _content(content) {}
std::string toString(); }; int main() { auto author = "Jane"; auto tite = "3
tips to learn English effectively"; auto content = "To learn English
effectively, you can try these 3 tips.\n" + "1. Practice on a daily basis.\n"
+ "2. Take time to understand difficult paragraphs" + "3. Try new things
everyday" ; Post p(author, title, content); std::cout << p.toString(); }
```

Output

```
Author: Jane Title: 3 tips to learn English effectively Content: To learn
English effectively, you can try these 3 tips. 1. Practice on a daily basis.
2. Take time to understand difficult paragraphs 3. Try new things everyday
```

Requirements

Create a **GraphPost** class, which represent a **Post** with more information.

- Add support for counting the number of **likes** for each post.
- Add support for a list of **comments** for each post. Each comment has the information of the commenter and the content.

You **should inherit** the class **GraphPost** from the class **Post**.

- Rewrite the above main function to create a **GraphPost** object and print out this GraphPost onto the console.

Sample output

```
Author: Jane Title: 3 tips to learn English effectively Content: To learn
English effectively, you can try these 3 tips. 1. Practice on a daily basis.
2. Take time to understand difficult paragraphs 3. Try new things everyday
Likes: 5 Comments: 3 - Anna: Great work. - Blash: This helps me a lot! -
Jack: Thanks for your post.
```